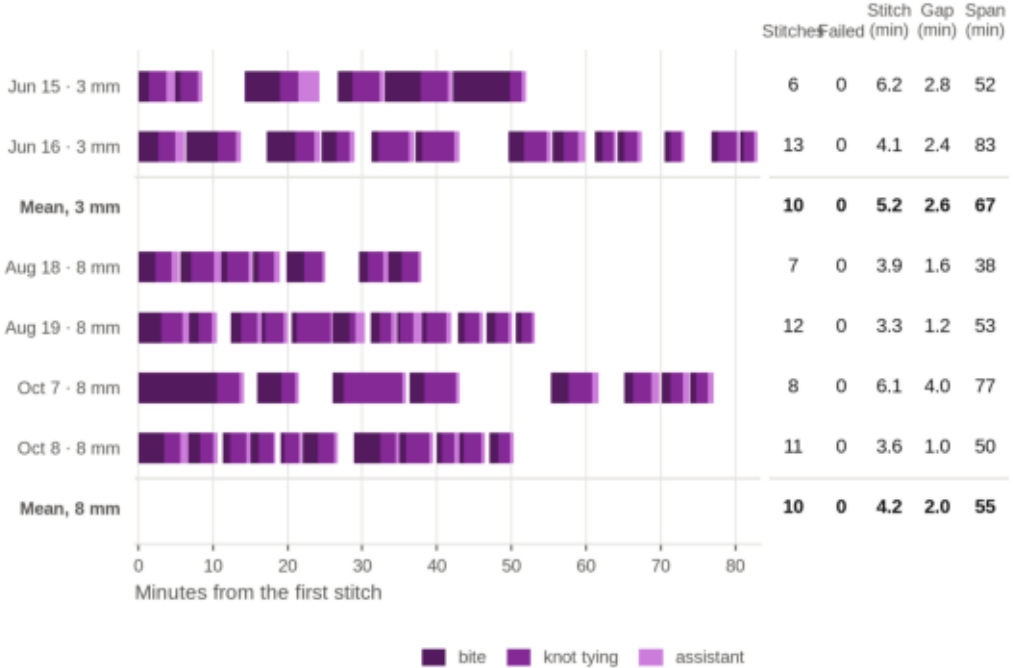
Pig	Knots tied	Failed attempts	Stitch (min)	Knot tying (min)	Assistant (s)	Force, mean (N, est.)	Force, peak (N, est.)
Jun 15 · 3 mm	6	0	6.2	2.7	59	1.06	2.22
Jun 16 · 3 mm	13	0	4.1	2.4	33	0.96	2.21
Mean, 3 mm	**10**	**0**	**5.2**	**2.5**	**46**	**1.01**	**2.22**
Aug 18 · 8 mm	7	0	3.9	2.4	34	1.39	3.72
Aug 19 · 8 mm	12	0	3.3	2.1	37	1.54	3.72
Oct 7 · 8 mm	8	0	6.1	3.2	38	1.74	4.15
Oct 8 · 8 mm	11	0	3.6	2.1	28	1.83	3.71
Mean, 8 mm	**10**	**0**	**4.2**	**2.5**	**34**	**1.63**	**3.82**

Knots tied: stitches with a completed knot. Stitch: needle first touches tissue to knot tightened. Knot tying: needle through to knot tightened. Assistant: knot tightened to the assistant's cut. Force (estimate, preliminary): insertion-axis joint effort minus a 60 s rolling median, mean and 95th percentile over each stitch, larger of the two arms; no force sensor. May 29 (3 mm) is excluded. Bold rows: mean of the pigs in each system.


Stitches
Failed
Stitch (min)
Gap (min)
Span (min)
Jun 15 · 3 mm
6 0 6.2 2.8 52
Jun 16 · 3 mm
13 0 4.1 2.4 83
Mean, 3 mm
10 0 5.2 2.6 67
Aug 18 · 8 mm
7 0 3.9 1.6 38
Aug 19 · 8 mm
12 0 3.3 1.2 53
Oct 7 · 8 mm
8 0 6.1 4.0 77
Oct 8 · 8 mm
11 0 3.6 1.0 50
Mean, 8 mm
10 0 4.2 2.0 55
0 10 20 30 40 50 60 70 80
Minutes from the first stitch
bite
knot tying
assistant